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A SEMI-SUPERVISED CLASSIFICATION MODEL

Figures 6 and 7 describe the architecture used in our experiments for semi-supervised classification (Section 4.3).

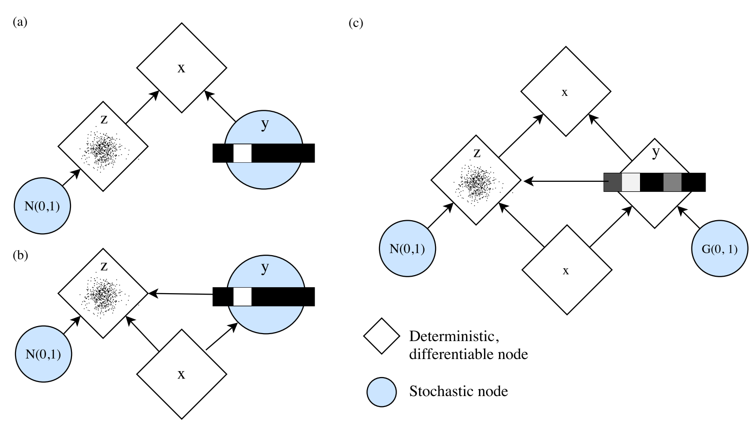

Figure 6: Semi-supervised generative model proposed by Kingma et al. (2014). (a) Generative model $p_\theta(x|y,z)$ synthesizes images from latent Gaussian "style" variable z and categorical class variable y. (b) Inference model $q_\phi(y,z|x)$ samples latent state y,z given x. Gaussian z can be differentiated with respect to its parameters because it is reparameterizable. In previous work, when y is not observed, training the VAE objective requires marginalizing over all values of y. (c) Gumbel-Softmax reparameterizes y so that backpropagation is also possible through y without encountering stochastic nodes.

B DERIVING THE DENSITY OF THE GUMBEL-SOFTMAX DISTRIBUTION

Here we derive the probability density function of the Gumbel-Softmax distribution with probabilities $\pi_1, ..., \pi_k$ and temperature τ. We first define the logits $x_i = \log \pi_i$, and Gumbel samples

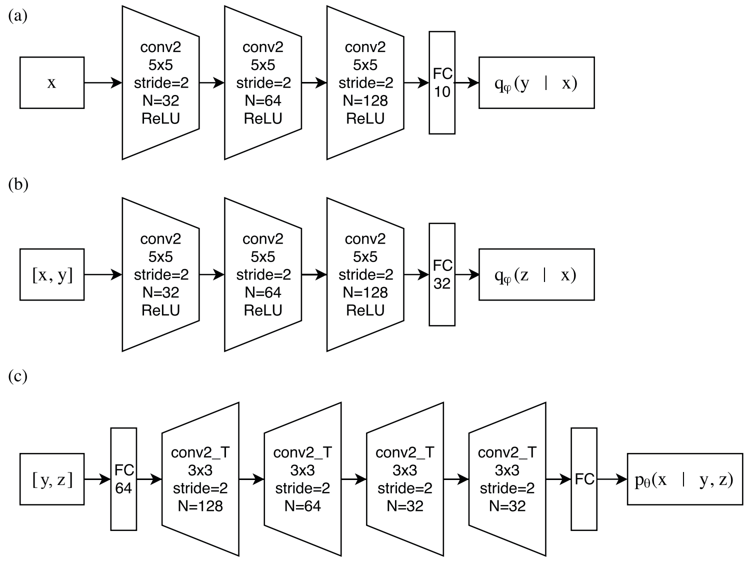

Figure 7: Network architecture for (a) classification $q_\phi(y|x)$ (b) inference $q_\phi(z|x,y)$, and (c) generative $p_\theta(x|y,z)$ models. The output of these networks parameterize Categorical, Gaussian, and Bernoulli distributions which we sample from.

$g_1, ..., g_k$, where $g_i \sim \text{Gumbel}(0, 1)$. A sample from the Gumbel-Softmax can then be computed as:

$$y_i = \frac{\exp\left((x_i + g_i)/\tau\right)}{\sum_{j=1}^k \exp\left((x_j + g_j)/\tau\right)} \qquad \text{for } i = 1, ..., k \tag{12}$$

B.1 Centered Gumbel density

The mapping from the Gumbel samples g to the Gumbel-Softmax sample y is not invertible as the normalization of the softmax operation removes one degree of freedom. To compensate for this, we define an equivalent sampling process that subtracts off the last element, $(x_k + g_k)/\tau$ before the softmax:

$$y_i = \frac{\exp\left((x_i + g_i - (x_k + g_k))/\tau\right)}{\sum_{j=1}^k \exp\left((x_j + g_j - (x_k + g_k))/\tau\right)} \qquad \text{for } i = 1, ..., k \tag{13}$$

To derive the density of this equivalent sampling process, we first derive the density for the "centered" multivariate Gumbel density corresponding to:

$$u_i = x_i + g_i - (x_k + g_k) \qquad \text{for } i = 1, ..., k-1 \tag{14}$$

where $g_i \sim \text{Gumbel}(0, 1)$. Note the probability density of a Gumbel distribution with scale parameter $\beta = 1$ and mean μ at z is: $f(z, \mu) = e^{\mu - z - e^{\mu - z}}$. We can now compute the density of this distribution by marginalizing out the last Gumbel sample, g_k:

$$\begin{aligned}
p(u_1, ..., u_{k-1}) &= \int_{-\infty}^{\infty} dg_k \, p(u_1, ..., u_k | g_k) p(g_k) \\
&= \int_{-\infty}^{\infty} dg_k \, p(g_k) \prod_{i=1}^{k-1} p(u_i | g_k) \\
&= \int_{-\infty}^{\infty} dg_k \, f(g_k, 0) \prod_{i=1}^{k-1} f(x_k + g_k, x_i - u_i) \\
&= \int_{-\infty}^{\infty} dg_k \, e^{-g_k - e^{-g_k}} \prod_{i=1}^{k-1} e^{x_i - u_i - x_k - g_k - e^{x_i - u_i - x_k - g_k}}
\end{aligned}$$

We perform a change of variables with $v = e^{-g_k}$, so $dv = -e^{-g_k} dg_k$ and $dg_k = -dv\, e^{g_k} = dv/v$, and define $u_k = 0$ to simplify notation:

$$p(u_1, ..., u_{k,-1}) = \delta(u_k = 0) \int_0^\infty dv\, \frac{1}{v} v e^{x_k - v} \prod_{i=1}^{k-1} v e^{x_i - u_i - x_k - v e^{x_i - u_i - x_k}} \tag{15}$$

$$= \exp\left(x_k + \sum_{i=1}^{k-1}(x_i - u_i)\right)\left(e^{x_k} + \sum_{i=1}^{k-1}\left(e^{x_i - u_i}\right)\right)^{-k} \Gamma(k) \tag{16}$$

$$= \Gamma(k) \exp\left(\sum_{i=1}^{k}(x_i - u_i)\right)\left(\sum_{i=1}^{k}\left(e^{x_i - u_i}\right)\right)^{-k} \tag{17}$$

$$= \Gamma(k)\left(\prod_{i=1}^{k} \exp\left(x_i - u_i\right)\right)\left(\sum_{i=1}^{k} \exp\left(x_i - u_i\right)\right)^{-k} \tag{18}$$

B.2 Transforming to a Gumbel-Softmax

Given samples $u_1, ..., u_{k,-1}$ from the centered Gumbel distribution, we can apply a deterministic transformation h to yield the first $k-1$ coordinates of the sample from the Gumbel-Softmax:

$$y_{1:k-1} = h(u_{1:k-1}), \qquad h_i(u_{1:k-1}) = \frac{\exp(u_i/\tau)}{1 + \sum_{j=1}^{k-1} \exp(u_j/\tau)} \forall i = 1, \ldots, k-1 \tag{19}$$

Note that the final coordinate probability y_k is fixed given the first $k-1$, as $\sum_{i=1}^{k} y_i = 1$:

$$y_k = \left(1 + \sum_{j=1}^{k-1} \exp(u_j/\tau)\right)^{-1} = 1 - \sum_{j=1}^{k-1} y_j \tag{20}$$

We can thus compute the probability of a sample from the Gumbel-Softmax using the change of variables formula on only the first $k-1$ variables:

$$p(y_{1:k}) = p\left(h^{-1}(y_{1:k-1})\right) \det\left(\frac{\partial h^{-1}(y_{1:k-1})}{\partial y_{1:k-1}}\right) \tag{21}$$

Thus we need to compute two more pieces: the inverse of h and its Jacobian determinant. The inverse of h is:

$$h^{-1}(y_{1:k-1}) = \tau \times \left(\log y_i - \log\left(1 - \sum_{j=1}^{k-1} y_j\right)\right) = \tau \times \left(\log y_i - \log y_k\right) \tag{22}$$

with Jacobian

$$\frac{\partial h^{-1}(y_{1:k-1})}{\partial y_{1:k-1}} = \tau \times \left(\mathrm{diag}\left(\frac{1}{y_{1:k-1}}\right) + \frac{1}{y_k}\right) = \begin{bmatrix} \frac{1}{y_1} + \frac{1}{y_k} & \frac{1}{y_k} & \cdots & \frac{1}{y_k} \\ \frac{1}{y_k} & \frac{1}{y_2} + \frac{1}{y_k} & \cdots & \frac{1}{y_k} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{1}{y_k} & \frac{1}{y_k} & \cdots & \frac{1}{y_{k-1}} + \frac{1}{y_k} \end{bmatrix} \tag{23}$$

Next, we compute the determinant of the Jacobian:

$$\det\left(\frac{\partial h^{-1}(y_{1:k-1})}{\partial y_{1:k-1}}\right) = \tau^{k-1} \det\left(\left(I + \frac{1}{y_k} e e^T \mathrm{diag}\left(y_{1:k-1}\right)\right)\left(\mathrm{diag}\left(\frac{1}{y_{1:k-1}}\right)\right)\right) \tag{24}$$

$$= \tau^{k-1}\left(1 + \frac{1 - y_k}{y_k}\right) \prod_{j=1}^{k-1} y_j^{-1} \tag{25}$$

$$= \tau^{k-1} \prod_{j=1}^{k} y_j^{-1} \tag{26}$$